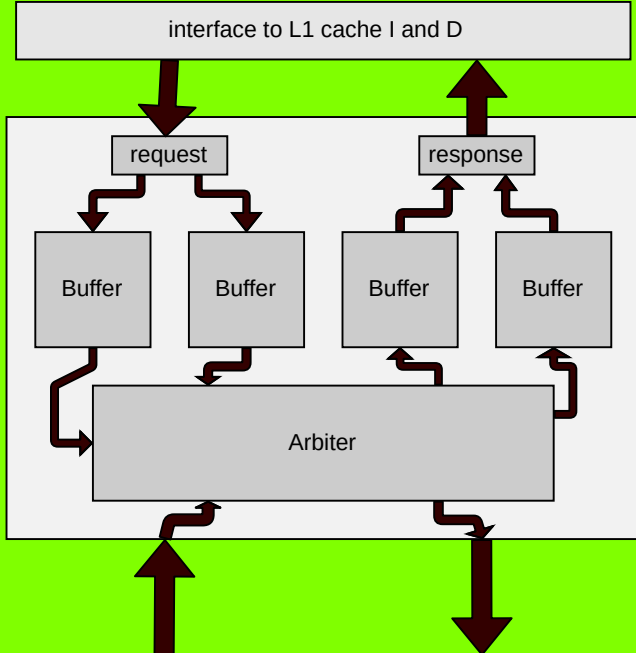
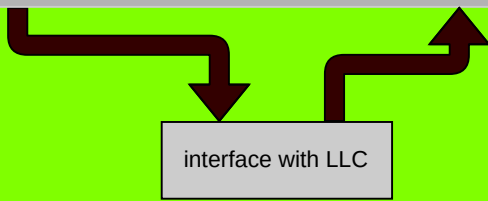
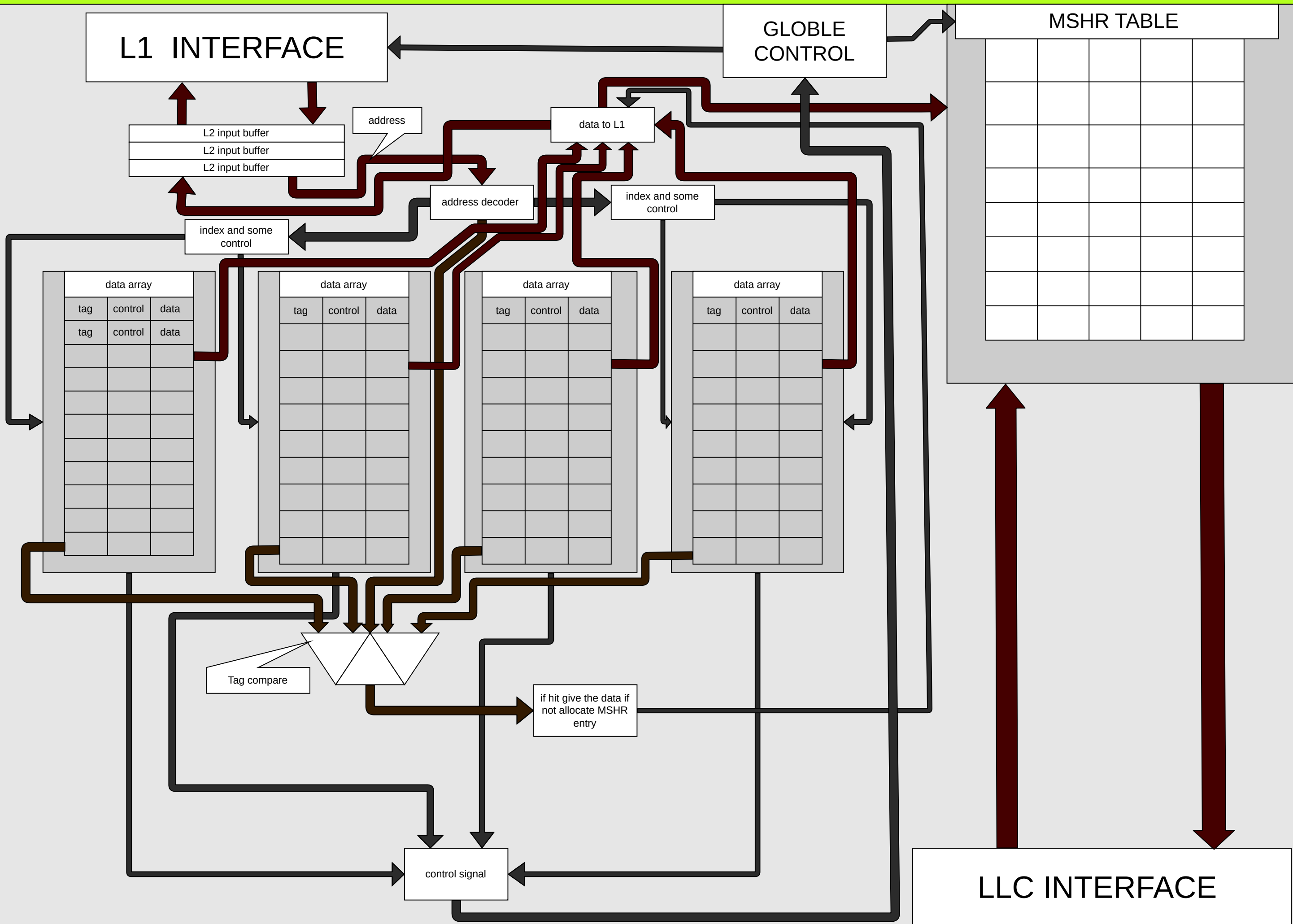




L2 cache memory





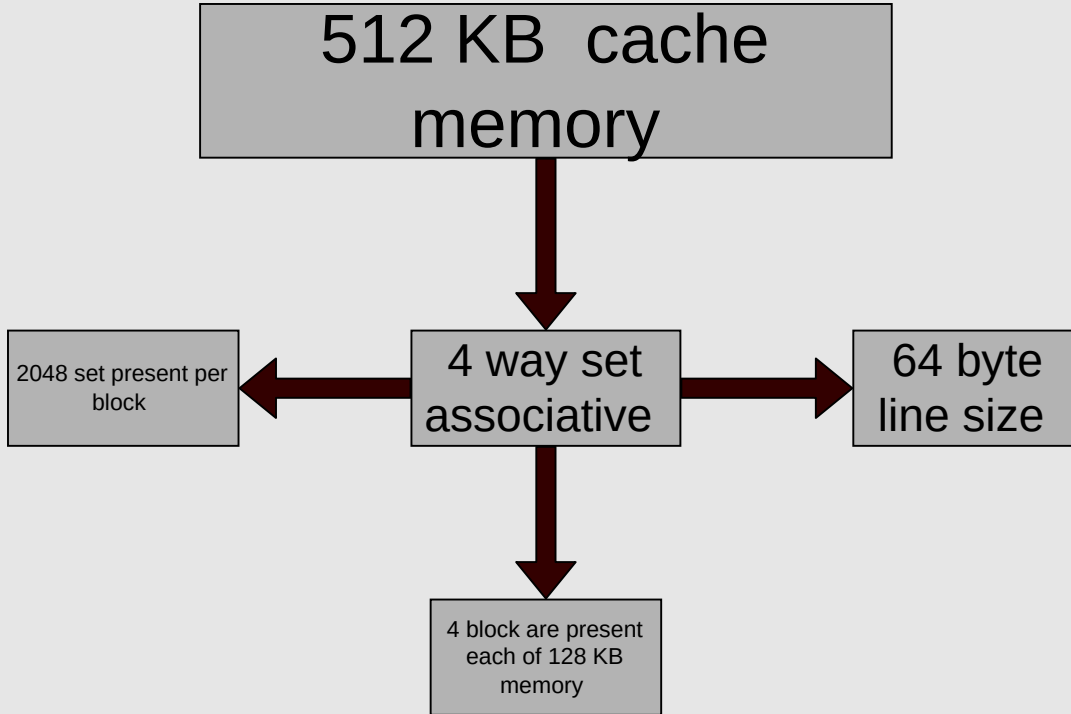
512 KB cache
memory

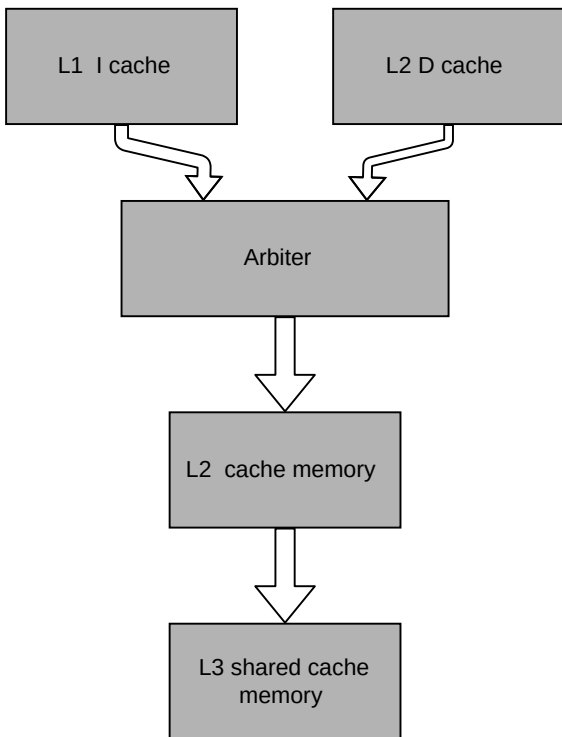
4 way set
associative

2048 set present per
block

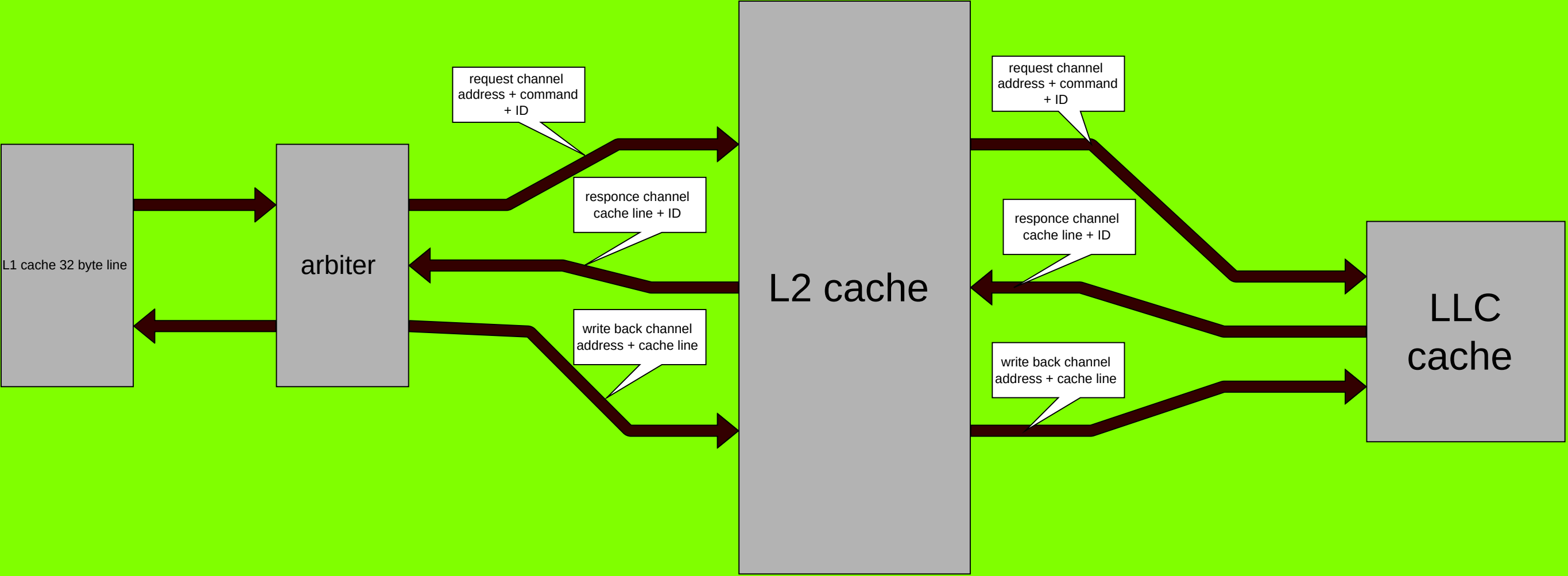
64 byte
line size

4 block are present
each of 128 KB
memory





DATA flow



input : index bit , hit ,
hit way , if miss hit = 0

output victm cache way

i am adding SRRIP replacement policy
(static re-refrence-interval prediction) this type of replacement
policy work very good for all type of workload

total extra 16384 bit
we need storage
overhead

SRRIP controler

Table

2 bit counter	2bit counter	2bit counter	2bit counter
way1	way2	way3	way4

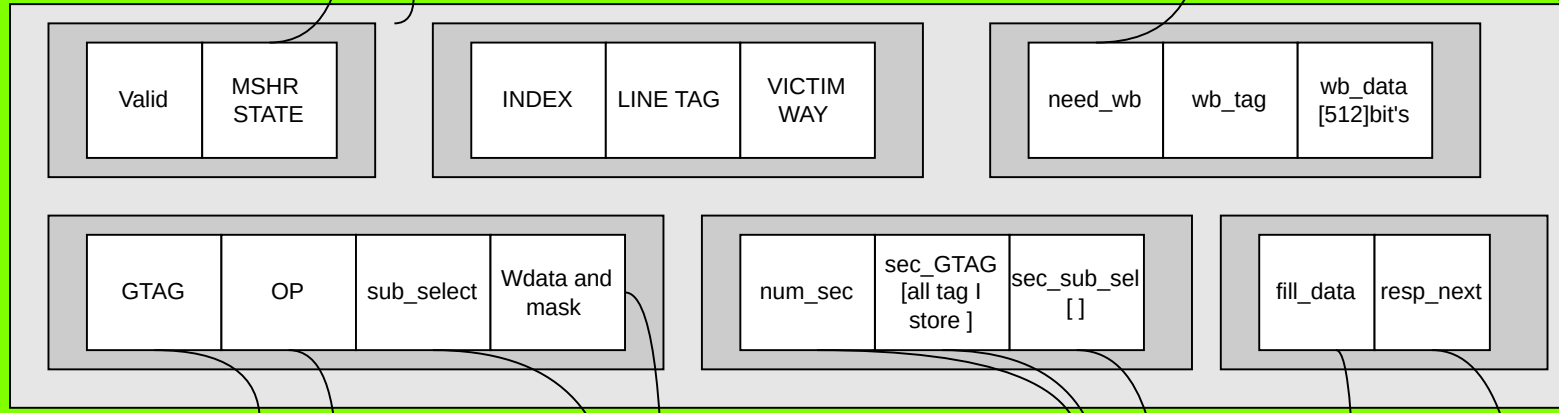
by the overhead of 8
bit per set , after
studing lot of
replacement policy I
decide to use this
i belive it work really
very good for all type
of workloads

MSHR STATE {
MSHR_IDLE
MSHR_WB_SEND
MSHR_WB_WAIT
MSHR_FILL_SEND
MSHR_FILL_WAIT
MSHR_COMMIT
MSHR_RESP }

each entry has there own FSM so 8 FSM which are control by GLOBLE FSM

If victim way is dirty we need to write it back

this is just one entry I decide to have such kind of 8 entries all the Entries compare at the same time



Globle transaction tag
(tag+sorce id)

store the operation that cause the miss

as the L1 memory has 32 byte lane so we need sub_select to select the L2 lane
1 = upper half and 0 = lower half

Store write data when write miss
byte enable mask

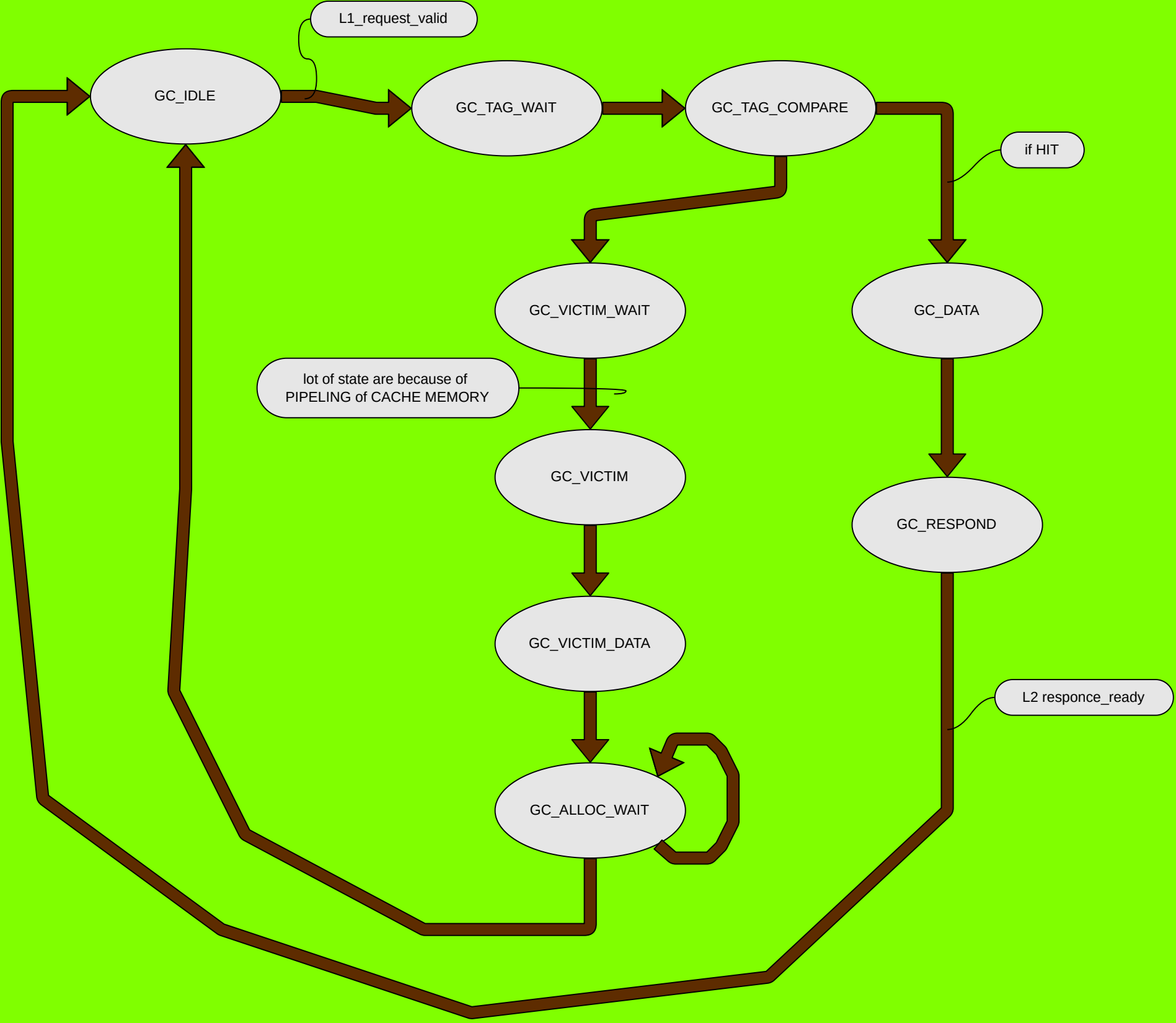
store the number of merge
secondary request
multiple request can be possible for
same data

store the GTAG of all merged
secondary request

sub_select for GTAG

Store the cache line return by the LLC

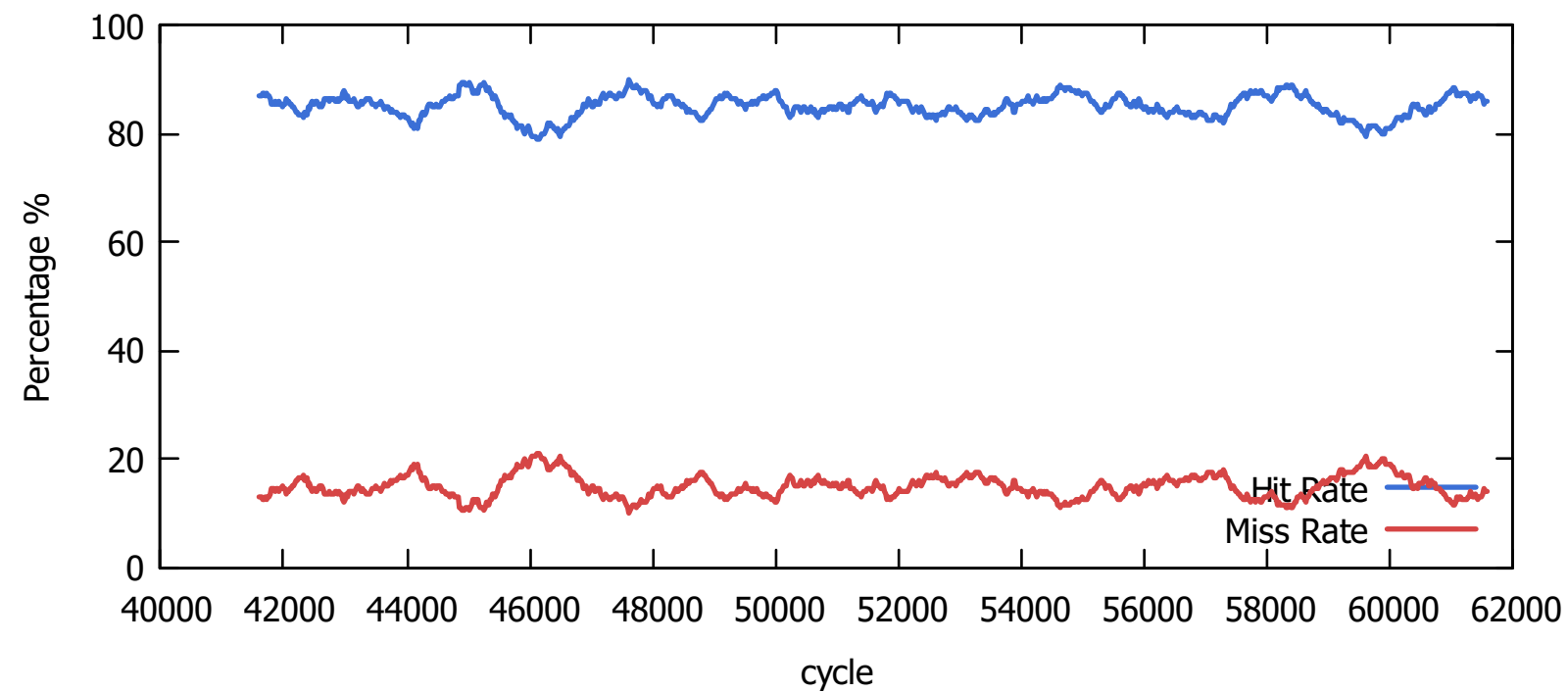
trach which requester should be the next responder



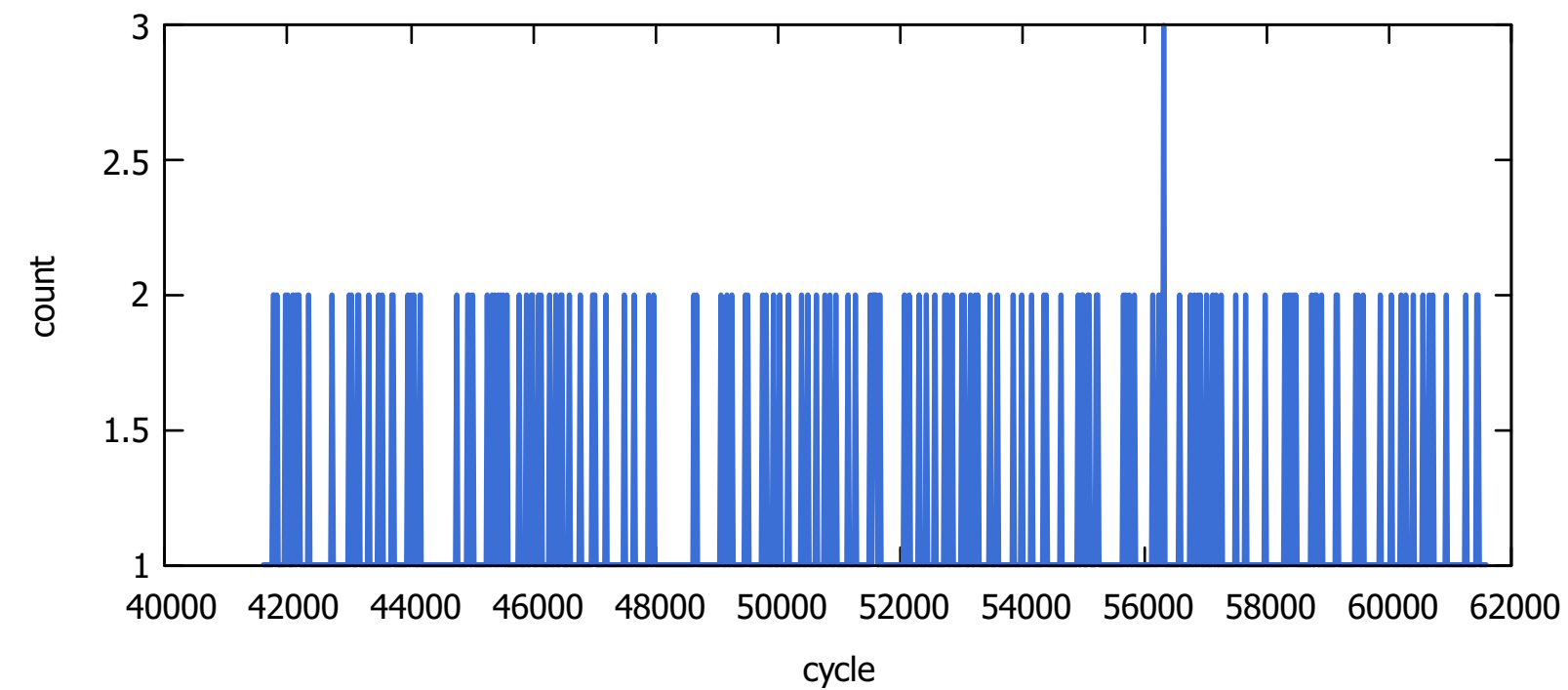
Globe control FSM

L2 Cache Performance Monitor

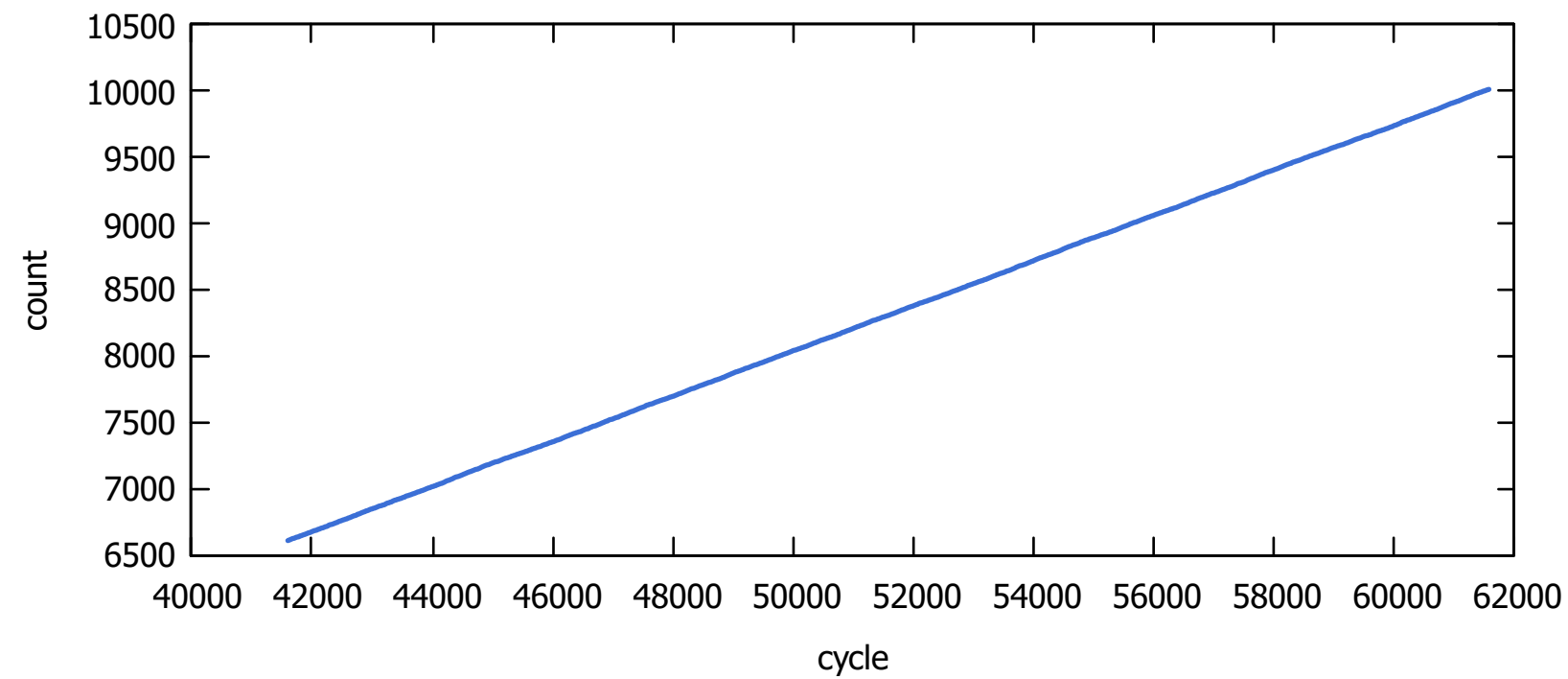
Hit Rate vs Miss Rate



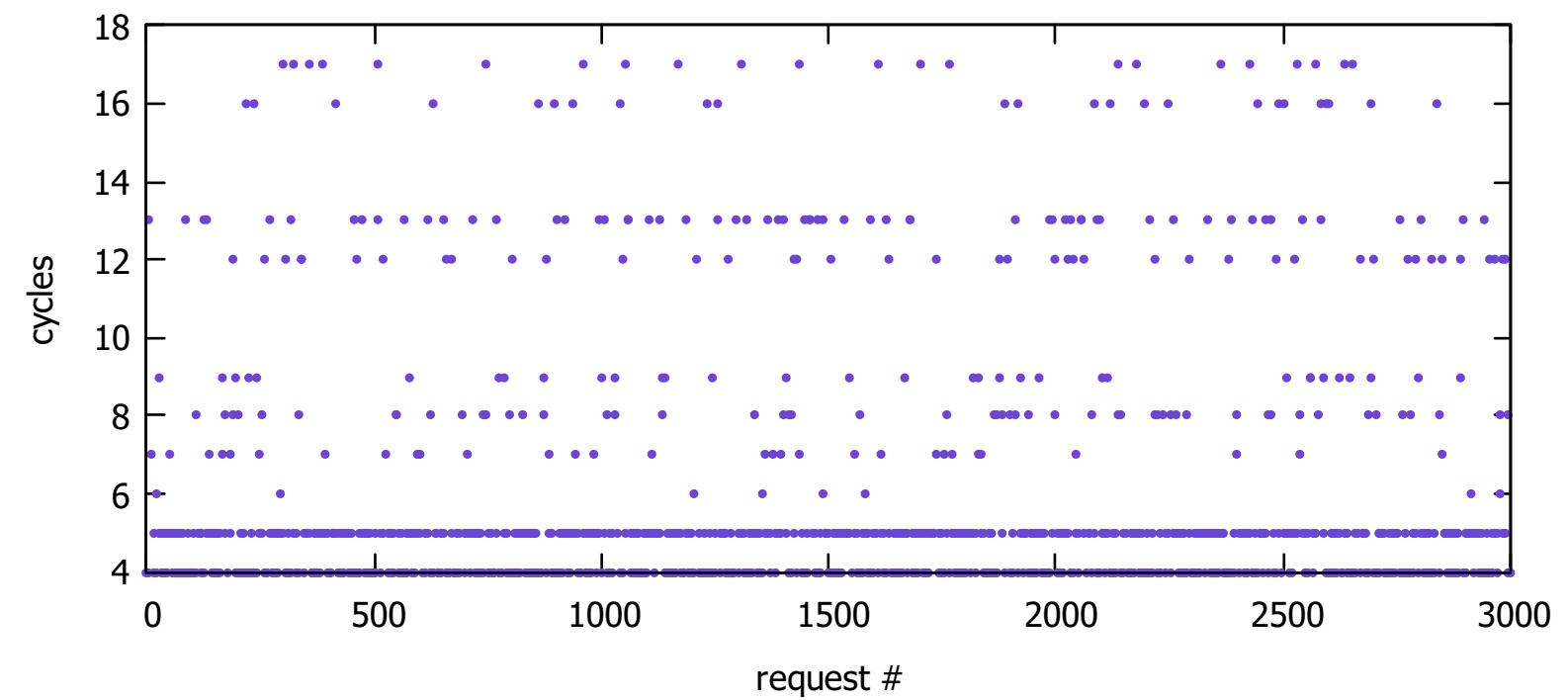
Outstanding requests (MSHR proxy)



Cumulative completed requests



Per-request latency



PROCESSOR TILE ORGANIZATION

